

Modelo de f.c. Integrales Diferenciales

$$v_e(t) = R_1 i_1(t) + \frac{1}{C} \int (i_1(t) - i_2(t)) dt$$

$$0 = \frac{1}{C} \int (i_2(t) - i_1(t)) dt + L \frac{di_2(t)}{dt} +$$

$$R_2 i_2(t) \rightarrow \frac{1}{C} \int (i_1(t) - i_2(t)) dt = L \frac{di_2(t)}{dt} + R_2 i_2(t)$$

Transformado de Laplace

$$v_e(s) = R_1 i_1(s) + \frac{1}{Cs} (i_1(s) - i_2(s))$$

$$\frac{1}{Cs} (i_1(s) - i_2(s)) = (Ls + R_2) i_2(s)$$

Sist. en Malla 1

$$v_e(s) = R_1 (Ls^2 + CR_2 s + 1) i_2(s) + (Ls + R_2) i_2(s)$$

$$v_e(s) - R_2 i_2(s) \rightarrow i_2(s) = \frac{v_e(s)}{R_2}$$

$$v_e(s) = \frac{(CLR_1)s^2 + (CR_2R_2 + L)s + (R_1 + R_2)}{R_2} v_e(s)$$

función de transferencia $V(s)$

$$\frac{V(s)}{V_d(s)} = \frac{R_2}{(CLR_1)s^2 + (CR_1R_2 + L)s + (R_1 + R_2)}$$

Error en estado estacionario

$$E(s) = V_d(s) - V(s)$$

$$E(s) = \frac{1}{s} (1 - FT)$$

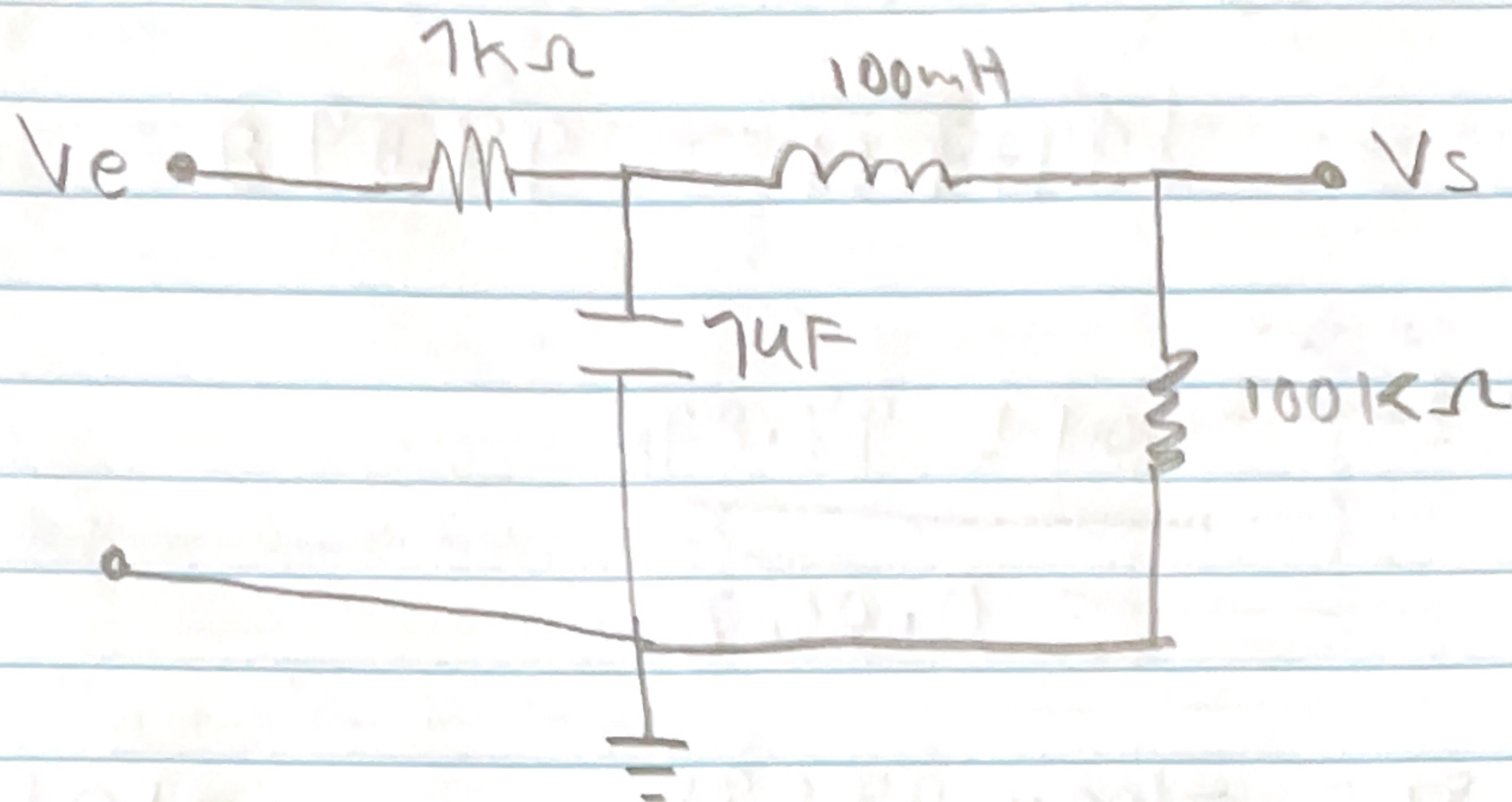
$$E_{ss} = \lim_{s \rightarrow 0} \left(\frac{R_2}{(CLR_1)s^2 + (CR_1R_2 + L)s + R_1 + R_2} \right)$$

$$e_{ss} = \frac{R_1}{R_1 + R_2}$$

$$e_{ss} \text{ control } = \frac{1k}{1k + 100k} = 0.0099$$

$$e_{ss} \text{ caso } = \frac{1k}{1k + 1k} = 0.5$$

Estabilidad del sistema Control



$$F.t. = \frac{100e^3}{[(1e^3)(100e^3)(1e^{-6})s^2 + (100e^{-3} + 1e^3(100e^3))((1e^{-6})s + (1e^3 + 100e^3))]}$$

$$= \frac{100,000}{0.0001 s^2 + 100.1 s + 101000}$$

$$as^2 + bs + c = 0$$

$$a = 0.0001$$

$$b = 100.1$$

$$c = 101000$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} ; \frac{-100.1 \pm \sqrt{100.1^2 - 4(0.0001)(101000)}}{2(0.0001)}$$

$$\underline{(101000)}$$

$$= (100.1)^2 = 10020.01$$

$$4(0.0001)(101000) = 40.4 \quad (1)$$

$$10020.01 - 40.4 = 9979.61$$

$$\sqrt{9979.61} = 99.898$$

$$S = \frac{-100.1 \pm 99.898}{0.0002}$$

$$S_1 = \frac{-100.1 + 99.898}{0.0002} = \underline{-1010}$$

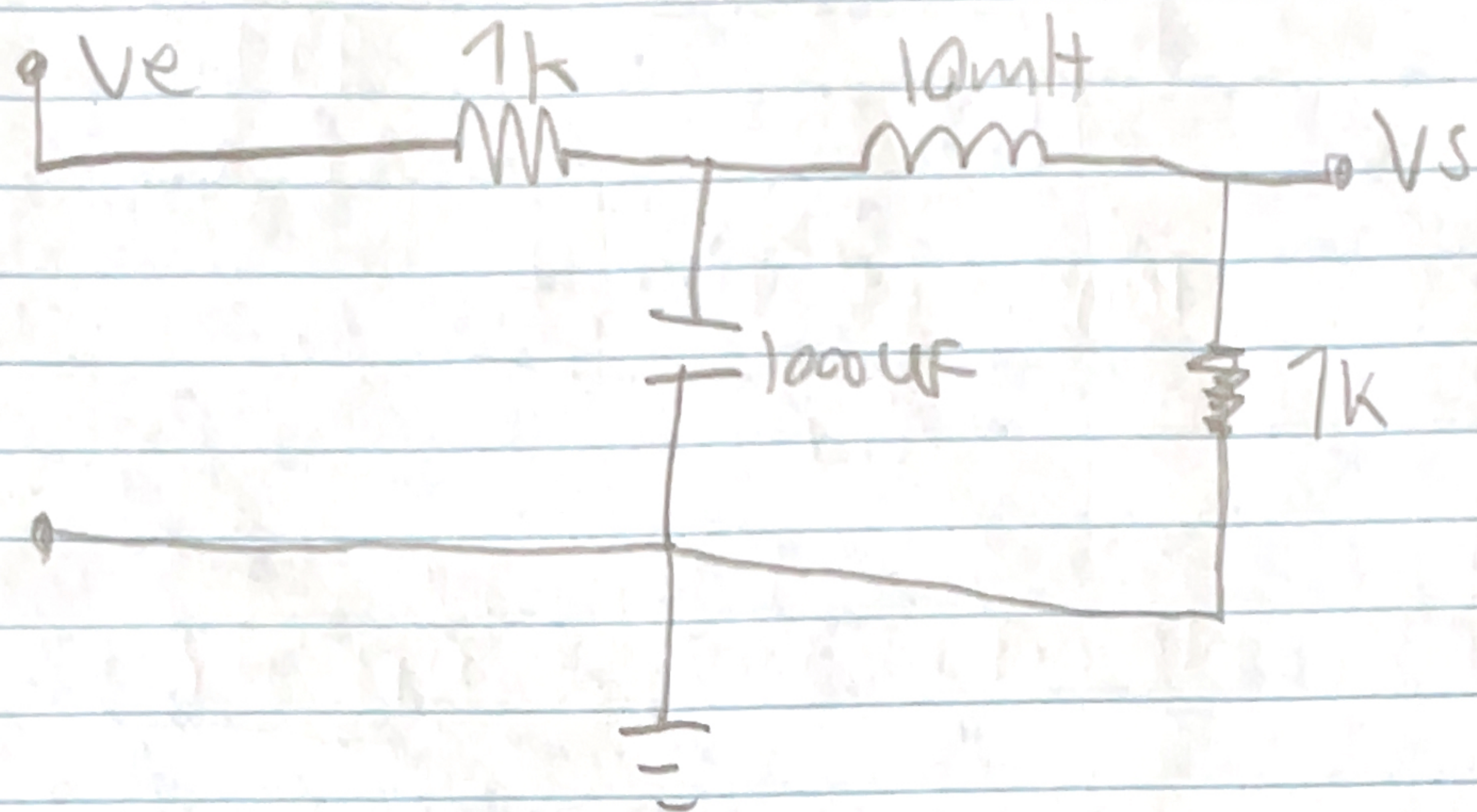
$$S_2 = \frac{-100.1 - 99.898}{0.0002} = \underline{S_2 = -999990}$$

* ambas raíces negativas

* el sistema control es estable

* sistema sobreamortiguado

Estabilidad del sistema Caso



$$F_t = 100e^3$$

$$[(1e^3)(100e^3)(1e^{-6})s^2 + (100e^3 + 1e^3(100e^3)1e^{-6})s + (1e^3 + 100e^3)]$$

$$F \cdot T: \frac{1000}{0.7s^2 + 1000.7 + 2000}$$

$$a = 0.7 \quad b = 1000.7 \quad c = 2000$$

$$s = \frac{-1000.7 \pm \sqrt{(1000.7)^2 - 4(0.7)(2000)}}{2(0.7)}$$

$$(1000.7)^2 = 1000200.01$$

$$4(0.7)(2000) = 800$$

(3)

$$1000200.01 - 800 = 999400.01$$

$$\sqrt{999400.01} = 999.7$$

$$s = \frac{-1000.1 \pm 999.7}{0.2}$$

$$s_1 = \frac{-1000.1 + 999.7}{0.2} \quad s_1 = -2$$

$$s_2 = \frac{-1000.1 - 999.7}{0.2} = s_2 = -9999$$

* el sistema caso es estable

* Ambas raíces negativas

* el Sistema es sobre amortiguado